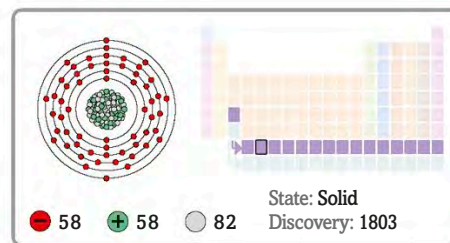


58
Ce

Cerium

Cerium was the first of the lanthanides to be discovered. It is named after the dwarf planet Ceres, which was discovered two years before the element was isolated. Cerium is highly toxic when **pure**, but safer cerium compounds have some uses. The main use of cerium is in making phosphors, which are chemicals that produce lights of different colours. Phosphors are present in **flatscreen TVs** and bulbs.

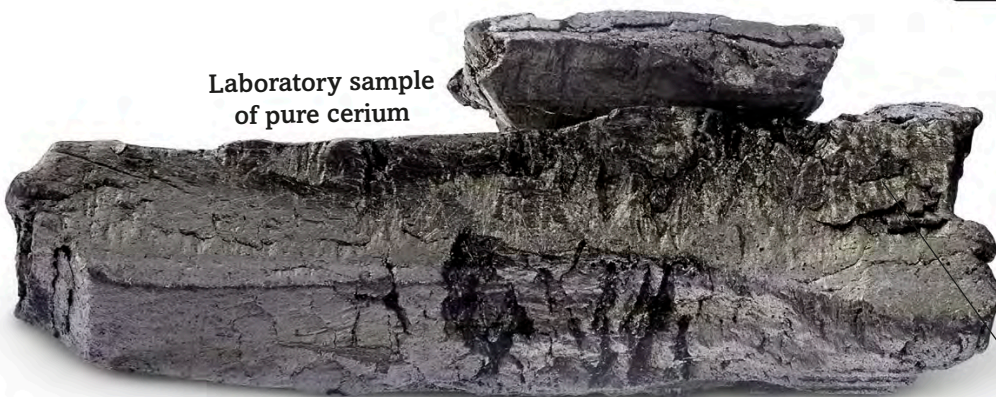


Television



The inside of this screen is coated with cerium-containing phosphors, which emit red, green, and blue light.

Laboratory sample of pure cerium



Kitchen spatula

This red colour comes from a compound called cerium sulfide.

The pure form of the metal tarnishes on contact with air.

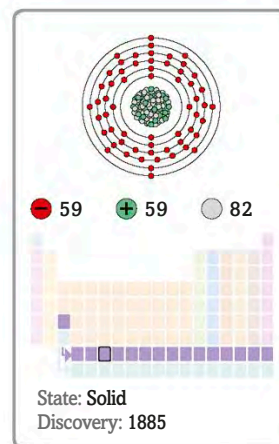
59
Pr

Praseodymium

This piece of the element's pure form is often stored in mineral oil to stop it from reacting with oxygen in the air.

Part of this element's name comes from *prasinos*, the Greek word for "green".

Normally a grey colour when pure, the element reacts slowly on contact with air to form a green coating. Praseodymium compounds give a yellow colour to glass and heat-resistant **ceramics**, and provide a green colour to some **artificial jewels**. This element also boosts the strength of magnets that contain it.

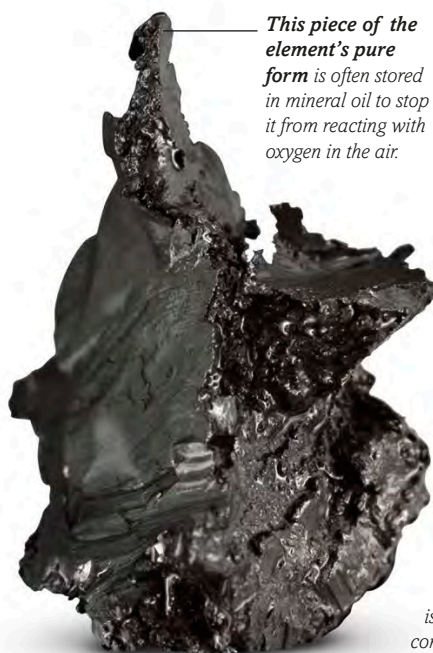


Yellow ceramic pot



This yellow colour is produced by a solution containing praseodymium.

Laboratory sample of pure praseodymium



Green cubic zirconia



This artificial gem gets its green colour from tiny amounts of a compound of praseodymium and oxygen.